

Real and Complex Analysis

1. GENERAL

- 1.1 TITLE::**Real and Complex Analysis
1.2 COURSE NUMBER (if known)::DC. MA 313.16
1.3 CREDITS::3-0-0 (9)
1.4 SEMESTER-OFFERED::Vth
1.5 PRE-REQUISITES::Engg Mathematics I & II
1.6 Course Committee Member:

2. OBJECTIVE::

The course aims to provide the students an indepth knowledge of real and complex functions concerning differentiation and integration.

3. COURSE TOPICS::

- UNIT I.** Metric spaces, Compactness, Connectedness (with emphasis on $\mathbb{R}^n$). (4 Lectures)
UNIT II. Measurable sets, Lebesgue measure and its properties, Product measure spaces, Fubini's theorem. (5 Lectures)
UNIT III. Simple and step functions, Lebesgue integral of step function, Upper functions, Lebesgue integral of upper functions, Lebesgue Integral functions, Fatou's Lemma, Dominated convergence Theorem, Monotone Convergence Theorem, Riemann integral as a Lebesgue integral, Lebesgue-vitali Theorem, Application of Lebesgue Integral. (9 Lectures)
UNIT IV. Jensen, Holder and Minkowski inequalities, Completeness, Duality of L^p and L^q for $\frac{1}{p} + \frac{1}{q} = 1$, Bounded functionals, Weak convergence on L^p , Uniform boundedness principal for $L^p(X, \mu)$, Approximation of L^p functions on $\mathbb{R}^n$ by smooth functions, Weak compactness of unit ball in L^p . (8 Lectures)
UNIT V. Zeros of holomorphic functions, Rouches theorem, Hurwitz's theorem, Maximum modulus principle, Open mapping theorem, Conformal mapping, Schwarz lemma, Normal families, Riemann mapping theorem. (6 Lectures)
UNIT VI. Harmonic function: Poisson integral formula, subharmonic functions, Dirichlet problem. (4 Lectures)

4. READINGS

4.1 TEXTBOOK::

1. H.L. Royden, Real Analysis, 3th ed., Macmillan.
2. W. Rudin, Real and Complex Analysis, McGraw-Hill.

4.2 *REFERENCE BOOKS::

1. G. de Barra, Measure Theory and Integration, New Age Publishers.
2. S. Ponnusamy, Foundations of Complex Analysis, Narosa Publishing House.

5. OTHER SESSIONS

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|-------------------|---|
| 5.1 *TUTORIALS:: | 0 |
| 5.2 *LABORATORY:: | 0 |
| 5.3 *PROJECT:: | 0 |

6. ASSESSMENT (indicative only)

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|------------------------|-----|
| 6.1 HA:: | 20% |
| 6.2 QUIZZES-HA:: | 0% |
| 6.3 PERIODICAL EXAMS:: | 40% |
| 6.4 *PROJECT:: | 0% |

6.5 FINAL EXAM:: 40%

7. OUTCOME OF THE COURSE:: Having done this course the student will be able to understand and adequately model physical and engineering problems.

8. *EXPECTED ENROLLMENT FOR THE COURSE:: 20